

From Primitives to Black Hole Architecture

A Walkthrough of the Event Density Black-Hole Derivation

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1 The Question

Black holes have produced the most contested set of paradoxes in modern theoretical physics. The information paradox, articulated by Hawking in 1975, asks whether quantum information that falls into a black hole is destroyed when the black hole evaporates — a possibility that would violate the unitarity of quantum mechanics. The firewall paradox of Almheiri, Marolf, Polchinski, and Sully (2012) asks whether an infalling observer must encounter a wall of high-energy radiation at the horizon, contradicting the equivalence principle’s prediction that a freely-falling observer should experience nothing locally unusual at the horizon. The complementarity contradiction, due to Susskind and others, attempts to resolve the information paradox by saying information is both inside and outside the horizon depending on observer, but runs into difficulties when entanglement monogamy is enforced at the geometric boundary.

These paradoxes have generated forty years of theoretical literature. They have not been resolved within standard theoretical frameworks. They are real puzzles, taken seriously by serious physicists, and the absence of consensus on their resolution is one of the central open problems in fundamental physics.

The Event Density framework’s contribution to black hole physics is structurally different from the standard paradox-resolution programs. The framework does not solve the information paradox. It does not resolve the firewall paradox. It does not arbitrate between complementarity and its alternatives. Instead, it shows that all of these paradoxes are generated by structural assumptions — specifically four of them — that the framework’s substrate ontology does not make. Without the assumptions that generate the paradoxes, the paradoxes do not arise.

The four assumptions are:

Global unitarity of the full quantum state on a single Hilbert space.

Global Cauchy data specifying the entire universe’s quantum state on a connected spacelike surface.

Geometric horizon as a sharp, primitive boundary in spacetime.

Monogamy of entanglement enforced at that geometric boundary.

Standard treatments impose all four. The information paradox arises from combining (i) and (iii): unitarity requires information preservation, but the geometric horizon seems to allow information to fall through and be destroyed. The firewall paradox arises from combining (i), (iii), and (iv): unitarity, the geometric boundary, and monogamy together force a high-energy wall at the horizon. Complementarity attempts to dissolve these by relaxing locality, but the geometric boundary in (iii) makes it hard to specify *where* information is, and monogamy in (iv) keeps producing contradictions.

The framework refuses all four assumptions at the substrate level. There is no global unitarity requirement — the framework is local and irreversible at the substrate level via P11 commitment irreversibility. There is no global Cauchy data — the framework commits events locally, event by

event. There is no geometric horizon as primitive — the horizon is a participation-bandwidth-collapse surface, not a smooth geometric boundary. There is no monogamy at boundary — the surface is not a knife-edge geometric locus, so there is nothing to enforce monogamy at.

The phrase that captures this: *black holes don't have paradoxes because black holes don't have smooth edges, because the substrate is discrete*. The smooth-horizon idealization that the standard paradoxes require simply isn't there. The substrate is discrete. The horizon is a statistical feature of bandwidth collapse, not a geometric primitive. The interior has no singularity because three substrate constraints jointly forbid divergence. The whole architecture rests on one substrate-level condition — a critical gradient threshold — that supports horizon formation, interior saturation, information blocking, entropy capacity, and scattering structure as different consequences of the same underlying mechanism.

This is a strong claim, and it deserves careful unpacking. What the framework asserts is structural completeness at the architectural level: every load-bearing piece of black-hole physics traces to substrate primitives plus the same critical-gradient condition. What it does not assert is full computational closure. The Hawking spectrum derivation matching $T_H = \kappa/(2\pi)$ is open — the mechanism is in hand (asymmetric participation at the saturated decoupling surface), but the explicit spectrum calculation requires future arc work. The Bekenstein–Hawking $1/4$ entropy coefficient is inherited rather than derived — the area-law form is forced, but the specific coefficient is set by substrate constants that aren't currently extracted in closed form. The framework operates within the acoustic-metric scalar-tensor approximation; full general relativity remains speculative rather than derived.

The chain has six steps:

Horizon as decoupling surface. Steep ED-gradients plus the V1 finite-width kernel plus DCGT multi-scale expansion produce exponential cross-bandwidth suppression. The decoupling surface forms where the suppression exceeds hydrodynamic-window resolution. Universalization across BH, Rindler, cosmological, and acoustic horizons is forced — same mechanism, different gradient sources.

No singularities. Three substrate constraints (finite participation via P4 + N1, discrete micro-events via P01, gradient saturation via P4 mobility-capacity bound) jointly forbid divergent curvature, energy density, tidal forces, and geodesic incompleteness. The acoustic-metric approximation breaks down at the same critical scale that defined the horizon, marking the boundary of where continuum description applies. The classical singular point is replaced by a finite-thickness saturated participation zone.

Information architecture. Information (committed structure) cannot cross the horizon by participation-bandwidth prohibition, not causal prohibition. Entanglement (uncommitted structure) can straddle the surface. Evaporation is participation re-routing through pair-creation-class events at the decoupling surface, with global correlation preserved. The four paradox-generating assumptions are explicitly not imposed at substrate level.

Entropy. Area-law form $S \propto A/\ell_P^2$ is forced from gradient saturation plus per-patch finite alphabet plus two-dimensional surface support. The $1/4$ coefficient is inherited rather than derived — the form is structural, the specific coefficient is a substrate constant currently outside closed-form extraction.

Scattering. BHPT phase shift $\delta_{\ell m} = \int \Delta \mathcal{A} d\lambda$ as global path integral of channel action. Helicity preservation under axisymmetric backgrounds, helicity flip under frame-dragging backgrounds, Kerr twist as integrated frame-dragging vorticity.

Architectural unification. One substrate-level condition $|\nabla \rho| \cdot \ell_P^2/\rho_{\text{local}} \gtrsim \log(R_{\text{cg}}/\ell_P)$ supports all six derivations. The architecture is internally consistent. The standard paradoxes are not generated.

The structural payoff: black-hole architecture in the framework is what falls out when the substrate’s discreteness, finite participation bandwidth, and commitment irreversibility are pushed through to the strong-gradient regime. The horizon is not a smooth geometric boundary. The interior is not a singular point. Information is not destroyed and is not paradoxically preserved — it is locally committed and bandwidth-blocked from crossing the decoupling surface. The whole architecture is parameter-free at the form level, with specific values inherited from substrate constants and one substrate condition supporting six different consequences.

2 The Substrate-Gravity Ontology

The framework rests on substrate-level ontological commitments. The black-hole arc uses the same substrate-gravity ontology that the Newton/MOND/BTFR walkthrough used, with three additional pieces that specifically matter for BH architecture: the V1 kernel, the P11 commitment-irreversibility primitive, and the DCGT bandwidth-suppression mechanism.

Micro-events and event density. Reality consists of discrete acts of becoming. The local rate of micro-event production at a region x is the event density $\rho(x)$. Higher ρ corresponds to faster local becoming; lower ρ to slower.

Participation and adjacency. Micro-events are linked by participation, the relational structure that determines whether one region can integrate the micro-events of another.

Chain stability landscape. A chain at substrate position x evaluates its next propagation step against the local participation environment via a stability landscape $\Sigma = \text{Coh} - \text{Str} - \text{Grad}$. The chain extends to the available next-state with maximum stability.

Decoupling surfaces. Surfaces where reciprocal participation between two regions becomes one-sided due to extreme ED-gradients. These will be the horizon-defining objects.

Participation-count bound. $N_{\text{dof}} = A/\ell_{\text{ED}}^2$ — the number of distinguishable participation degrees of freedom on a surface of area A . Forced by substrate-level UV-finiteness plus event-discreteness.

Three additional pieces enter the BH arc:

V1 finite-width kernel. From Theorem N1, the vacuum-response kernel V1 has finite spatial width $\sim \ell_P$. This is not a delta function — the substrate has a discreteness scale, and vacuum response operates at that scale rather than at arbitrarily-fine resolution. The finite-width property is what produces exponential cross-bandwidth suppression in the steep-gradient regime.

P11 commitment irreversibility. Every commitment event is irreversible at the substrate level. Once a participation event commits, the commitment cannot be reversed. This is the load-bearing primitive for both the kernel-level arrow of time (separate walkthrough) and the information architecture of black holes.

DCGT machinery. The Diffusion Coarse-Graining Theorem provides the substrate-to-continuum bridge: hydrodynamic-window scale separation $\ell_P \ll R_{\text{cg}} \ll L_{\text{flow}}$, multi-scale expansion of substrate flux statistics, form-FORCED / value-INHERITED structural split. DCGT is what allows the framework to talk about both substrate-level structure (where black-hole physics actually happens) and continuum-level structure (where the standard GR description applies).

The acoustic-metric framework. From the framework’s curvature-emergence arc (Arc ED-10), the substrate participation density modulates an effective acoustic metric $g_{\mu\nu}^{\text{acoustic}}$ that propagates substrate-level signals. This is a kinematic summary of substrate participation density rather than a fundamental dynamical field. The framework’s gravity content operates within this acoustic-metric scalar-tensor approximation; full GR remains speculative.

A note on what the framework explicitly does not claim. It does not derive the Einstein equation $G_{\mu\nu} = 8\pi GT_{\mu\nu}$ as a fundamental dynamical equation. It does not produce a quantum theory of gravity in the sense of quantizing the metric. It works within the acoustic-metric kinematic framework — substrate-level participation density modulates an effective metric, but the metric is not a fundamental field. This is the ED-Phys-10 guardrail that the BH arc preserves throughout.

That’s the working set. The horizon mechanism, singularity dissolution, information architecture, entropy, and scattering all operate on this substrate ontology.

3 The Horizon as a Decoupling Surface

The first structural result is the substrate-level mechanism that produces what standard GR calls an event horizon. The argument runs through three steps: gradient steepening produces participation anisotropy, the V1 finite-width kernel suppresses cross-surface commitment, and DCGT multi-scale expansion identifies the surface where the suppression exceeds hydrodynamic-window resolution.

3.1 Gradient steepening and participation anisotropy

Consider two adjacent substrate regions R_- and R_+ separated by a boundary surface Σ , with cell-averaged participation densities ρ_- and ρ_+ . The V1-mediated chain-step transition rate from R_- to a substrate site at displacement δ into R_+ is:

$$\Gamma(\delta; \rho_-, \rho_+) = \Gamma_0(\rho_{\text{local}}) \cdot K_{V1}(|\delta|)$$

where Γ_0 is the local participation-bandwidth-modulated rate and K_{V1} is the V1 finite-width spatial kernel.

When the ED-gradient $\nabla\rho$ across Σ is small, ρ_- and ρ_+ are similar, Γ_0 is comparable on both sides, and chain-step transitions in both directions are roughly balanced. This is the standard substrate regime where DCGT scalar-diffusion produces the canonical continuum-scale flux $\nabla \cdot (M(\rho)\nabla\rho)$.

When the gradient becomes steep — specifically when $|\nabla\rho| \cdot \ell_P / \rho_{\text{local}} \gtrsim 1$ — the local participation environment of a chain near Σ is dominated by the gradient direction. The chain’s available next-states are no longer isotropic; they are biased along the gradient. This is local participation anisotropy, and it is the substrate-level analog of strong-curvature in GR.

3.2 V1 kernel suppresses cross-surface commitment

The V1 kernel has finite spatial width $\sim \ell_P$ (Theorem N1). For chain-step transitions across Σ , the V1 kernel weights paths by their substrate displacement. In the regime where ρ varies across the kernel’s width — when $|\Delta\rho|_{\text{across-width}} \sim |\nabla\rho| \cdot \ell_P \gtrsim \rho_{\text{local}}$ — the kernel’s effective support across Σ is asymmetric.

The mechanism: $\Gamma_0(\rho)$ is a P4-class participation-bandwidth-modulated transition rate. At high ρ , Γ_0 saturates toward zero (the P4 mobility-capacity bound). A chain on the high- ρ side near Σ sees its forward (toward higher ρ) and backward (toward lower ρ) transition rates differ by a factor $\Gamma_0(\rho_+)/\Gamma_0(\rho_-)$. As $\rho_+ \rightarrow \rho_{\text{max}}$ (P4 saturation), the forward rate goes to zero; the backward rate is suppressed by an effective potential barrier from the steep gradient.

Quantitatively, the V1-mediated transition rate across Σ at gradient scale $|\nabla\rho|$ takes the form:

$$\Gamma_{\text{cross}}(\nabla\rho) \sim \Gamma_0(\rho_{\text{local}}) \cdot K_{V1}(\ell_P) \cdot \exp\left[-\alpha |\nabla\rho| \ell_P^2 / \rho_{\text{local}}\right]$$

with α a dimensionless V1-profile coefficient inherited at value layer (set by the kernel shape). The exponential suppression is forced by the multi-scale expansion: substrate displacements crossing the gradient at scale ℓ_P accumulate exponential phase-decoherence in the V1-mediated chain-step propagation.

When $|\nabla\rho| \cdot \ell_P^2/\rho_{\text{local}} \gg 1$ — the steep-gradient regime — the cross-participation bandwidth Γ_{cross} is exponentially suppressed.

3.3 The decoupling surface

Apply DCGT multi-scale expansion to the participation flux across Σ . The cell-averaged flux across the boundary becomes exponentially suppressed in the steep-gradient regime. There exists a critical gradient scale $|\nabla\rho|_c$ above which the flux is below the DCGT hydrodynamic-window threshold — meaning the substrate-to-continuum coarse-graining at scale R_{cg} cannot resolve any nonzero cross-participation bandwidth across Σ .

The decoupling surface Σ_{decouple} is defined formally:

$$\Sigma_{\text{decouple}} \equiv \{x : |\nabla\rho(x)| \cdot \ell_P^2/\rho_{\text{local}} \gtrsim \log(R_{\text{cg}}/\ell_P)\}$$

The logarithmic factor $\log(R_{\text{cg}}/\ell_P)$ is set by the DCGT hydrodynamic-window scale separation: cross-participation bandwidth is below DCGT resolution when the exponential suppression exceeds the dynamic range of the multi-scale expansion.

This is the substrate-derivation of horizon-as-decoupling-surface. The horizon is not a smooth geometric boundary. It is a statistical feature of the participation network — the surface where bandwidth collapse exceeds the resolution of the substrate-to-continuum coarse-graining.

3.4 Observer-independence at substrate level

A natural concern: an external observer sees a chain freezing and redshifting at the horizon, but the infalling chain experiences nothing pathological locally. How can both descriptions be consistent?

The DCGT-grounded version of this reconciliation:

The infaller's local ρ near the chain is smooth and finite. The chain's local participation environment is well-defined; cell-averaging at scale R_{cg} produces canonical DCGT continuum dynamics. The chain experiences nothing pathological at Σ_{decouple} because the surface is not a feature of the chain's local participation environment.

The external observer's cross-participation flux from the chain (now on the high- ρ side of Σ) to the external observer (on the low- ρ side) is exponentially suppressed via §3.2. The external observer cannot integrate the chain's micro-events at the rate the chain produces them. The chain appears to freeze, fade, redshift.

Both descriptions arise from the same ED gradient. The location of Σ_{decouple} is determined by the gradient — not by either observer's frame. The surface is observer-independent at the ontological level (the gradient is what it is) and observer-dependent only at the level of participation (which side of the gradient the observer is on determines what they experience).

3.5 Universalization across horizon types

The substrate mechanism is generic — it depends only on the existence of a steep ED gradient, not on the source of that gradient. Different physical situations produce steep ED gradients via different mechanisms:

Black hole horizons. A localized mass M produces a steep ED-gradient via the cumulative-strain mechanism of T19 (the Newton’s- G derivation in the gravity walkthrough). At the radius where the gradient reaches the critical scale, a decoupling surface forms. This radius corresponds approximately to the standard Schwarzschild radius $r_S = 2GM/c^2$ in the weak-field limit, with substrate-cutoff corrections of order ℓ_P^2/r_S^2 (negligible at astrophysical scales).

Rindler horizons. A uniformly accelerating chain breaks 3D participation isotropy along the acceleration direction; the acceleration-induced decoupling surface lies at distance $d_a = c^2/a$ behind the chain. The substrate mechanism is identical — the chain’s acceleration produces a participation-density gradient that, at distance d_a , exceeds the critical scale.

Cosmological horizons. The cosmic horizon at $R_H = c/H_0$ is the participation-density boundary of the universe at cosmological scale (the same R_H that appeared in the MOND transition acceleration derivation in the gravity walkthrough). The Hubble-rate H_0 produces a substrate-level participation-density gradient at scale R_H .

Acoustic and analog horizons. In analog-gravity systems (transonic fluid flow, BECs with sonic horizons, optical analog systems), the participation-density analog is the local fluid density, refractive index, etc. The mechanism is identical: when the gradient of the analog-density variable exceeds a critical scale, a decoupling surface forms for the relevant participation-mediated signals.

All four horizon types are structurally identical under DCGT. Each is a surface where cross-participation bandwidth collapses below the DCGT hydrodynamic-window resolution. The unification is forced by the universalized mechanism — same bandwidth-collapse mechanism, different gradient sources.

This is one of the cleanest universalization results in the framework. It explains why analog gravity reproduces horizon-like phenomena despite involving no actual GR curvature: the substrate mechanism that produces horizons doesn’t require GR curvature, only steep gradients in the participation-density analog.

3.6 The geometric horizon as continuum artifact

How does the substrate decoupling surface relate to the standard GR notion of an event horizon?

Under the acoustic-metric reading of Arc ED-10, the substrate participation density modulates an effective metric $g_{\mu\nu}^{\text{acoustic}}$ that propagates substrate-level signals. At a decoupling surface Σ_{decouple} , the cross-participation bandwidth is exponentially suppressed — meaning substrate-mediated signal propagation across Σ is suppressed.

Translated to the acoustic-metric continuum description: the effective null cone tilts so that signal-cone-future-of- Σ no longer intersects the external region. This is the standard GR statement of “null surface separating causally connected from causally disconnected regions” — but in the substrate framework, the null surface is the continuum-emergent shadow of the underlying participation-bandwidth-collapse, not a fundamental geometric object.

Three structural consequences of this reframing:

Horizon teleology dissolved. Standard GR’s “global future” definition of event horizons creates a teleological-feeling structure (the horizon “knows” about its own future). In the substrate framework, the decoupling surface is local and immediate; teleological feeling is an artifact of continuum coarse-graining.

Horizon dynamics are participation-network reconfiguration. When matter falls into a BH and the horizon grows, what’s happening at substrate level is participation-network reconfiguration — the boundary of the bandwidth-collapse region expands as more substrate participation density becomes part of the high- ρ region.

No global Cauchy data needed. Standard GR’s event horizon definition requires global Cauchy data. Substrate-level decoupling surfaces require only local gradient configuration; global Cauchy data is a coarse-graining artifact.

The geometric horizon of standard GR is the continuum-emergent shadow of the substrate decoupling surface. The substrate has only the bandwidth-collapse surface, which is a statistical feature of the participation network rather than a smooth geometric boundary. This is the first piece of the no-paradoxes-because-no-smooth-edges framing: the horizon at substrate level isn’t a smooth geometric boundary that the standard paradoxes assume.

4 No Singularities

The second structural result is the substrate-level forbiddance of singularities. The argument runs through three substrate constraints that jointly forbid divergence, plus a strong-curvature audit identifying what specifically replaces the classical singular point.

4.1 Three substrate constraints forbidding divergence

Constraint 1 — Finite participation. The substrate’s participation bandwidth is finite. Per P4 plus Theorem N1, every chain-step transition has a finite rate $\Gamma_0(\rho) \leq \Gamma_{\max}$, with the rate going to zero as $\rho \rightarrow \rho_{\max}$. There is no substrate configuration where a chain can produce an unbounded number of commitment events in a bounded proper-time interval. This constraint forbids divergent commitment density. There is no $\rho \rightarrow \infty$ regime in the framework.

Constraint 2 — Discrete micro-events. Per Primitive 01 (event discreteness), substrate-level reality consists of discrete micro-events. There is no continuum point that can host infinite structural detail. A “point” in the continuum description corresponds, at substrate level, to a finite collection of micro-events with finite participation bandwidth. The notion of an infinity of micro-events compressed to a single point is ill-defined at substrate level. This constraint forbids infinitely-localized structure.

Constraint 3 — Gradient saturation. Per the P4 mobility-capacity bound, ED gradients saturate when local participation density approaches $\rho_{\max}$. The chain-step transition rate $\Gamma_0(\rho)$ vanishes as $\rho \rightarrow \rho_{\max}$. The substrate’s ability to sustain gradients structurally fails as participation density approaches saturation. There is a maximum-viable $|\nabla\rho|_{\text{viable}}$ beyond which the substrate physically cannot support further steepening — the gradient saturates structurally.

Jointly, these three constraints forbid the standard GR singularity-class pathologies:

Divergent curvature would require divergent gradients at substrate level. Forbidden by Constraint 3.

Divergent energy density would require divergent commitment density. Forbidden by Constraint 1.

Divergent tidal forces would require divergent gradient differences across small displacements. Forbidden by Constraints 2 and 3 jointly.

Divergent geodesic incompleteness in standard GR means certain timelike or null geodesics terminate after finite proper time at the singular point. At substrate level, “geodesics” are stable participation pathways. Stable participation pathways do not terminate at substrate-discrete-and-finite participation density; they continue (as substrate-level chain-step propagation) into the saturated region. The geodesic-incompleteness pathology is a continuum artifact.

The substrate-level claim is therefore: all singularity pathologies are continuum-coarse-graining artifacts that fail at the substrate scale. This grounds “ED predicts no singularities” in three explicit primitive-level constraints.

4.2 Strong-curvature audit: where the acoustic metric breaks down

The framework operates within the acoustic-metric scalar-tensor approximation. This approximation is valid under condition (C5) of Arc ED-10: the acoustic metric must remain Lorentzian and non-degenerate. The strong-curvature audit asks whether (C5) holds near a would-be singularity.

The acoustic metric is the participation-density-modulated effective metric — a kinematic summary of substrate participation density. The metric tensor’s components are functions of ρ and its gradients. For ρ well below $\rho_{\max}$ and gradients moderate, the metric is well-defined Lorentzian.

When local ρ approaches $\rho_{\max}$, participation-density saturation makes the substrate’s mobility coefficient vanish. The acoustic metric becomes degenerate — its determinant approaches zero, its signature ceases to be Lorentzian in a well-defined sense.

Define the dimensionless saturation parameter:

$$\sigma(x) \equiv |\nabla \rho(x)| \cdot \ell_P^2 / \rho_{\text{local}}$$

For $\sigma \ll 1$: weak-gradient regime, acoustic metric well-defined Lorentzian.

For $\sigma \sim 1$: transition regime, acoustic metric still Lorentzian but with substrate-cutoff corrections becoming non-negligible.

For $\sigma \gtrsim \beta_{\text{crit}}$ (where $\beta_{\text{crit}} \sim \log(R_{\text{cg}}/\ell_P)$ — the same critical scale that produced horizon formation): strong-gradient regime. The acoustic metric becomes degenerate. Condition (C5) fails.

The interior of a black hole, classically, has steeply rising curvature toward $r = 0$. At substrate level, this corresponds to participation density ρ rising toward $\rho_{\max}$, with σ entering the strong-gradient regime $\sigma \gtrsim \beta_{\text{crit}}$. Inside this regime, the acoustic-metric scalar-tensor approximation is no longer valid.

What the breakdown means: not that substrate dynamics fail. The substrate participation network is still well-defined; chains still commit; the V1 kernel still mediates transitions; DCGT still operates within its hydrodynamic-window scope. What fails is the coarse-grained continuum description — the acoustic-metric reading that produces the geometric language of GR-class physics.

This is structurally distinct from a coordinate-only breakdown like the Schwarzschild horizon (curable by Eddington–Finkelstein coordinates). The “true singularity” of standard GR corresponds to a manifold-model breakdown — the continuum approximation itself fails because participation-density saturation makes the metric degenerate, and no continuum coordinate change cures this.

4.3 The substrate replacement: a saturated participation zone

What replaces the classical singular point? In the interior region where $\sigma \gtrsim \beta_{\text{crit}}$, the substrate is in the participation-density-saturated regime. ρ is approaching $\rho_{\max}$, the mobility $\Gamma_0(\rho)$ is approaching zero, and substrate-level participation is highly constrained.

This is not a region where physics fails. It is a region where:

Substrate micro-events still occur, but at a saturating rate.

V1-mediated chain transitions are increasingly suppressed.

Cross-participation bandwidth has collapsed in all directions, not just radially outward — making this different from the horizon-decoupling, which collapses bandwidth in only one direction.

The participation network has become maximally committed — most available substrate participation channels are at or near saturation.

Within the saturated zone, local ED dynamics become effectively one-directional. The substrate-level forward-cone-only support of V1 (Theorem 18, the kernel-level arrow of time) is preserved, but spatial directions become structurally constrained: there is no substrate-meaningful “outward” direction (nothing can propagate outward through the saturated region) and no substrate-meaningful “transverse” direction (transverse chain-step transitions are also suppressed).

The remaining structural direction is the commitment-order direction (proper-time) of T18. Substrate dynamics within the saturated zone proceed forward in commitment order, but the spatial-direction information that would have made the dynamics geometrically interpretable has collapsed.

This is the substrate-level analog of the standard GR statement that “inside a Schwarzschild horizon, the radial direction becomes timelike.” At substrate level, all spatial directions become structurally ill-defined inside the saturated zone; only commitment-order remains as a meaningful direction.

The classical singular point is replaced by:

A finite-thickness saturated participation zone — not a point. Its size is set by $\rho_{\max}$ plus the substrate gradient profile plus the BH mass.

Not a curvature divergence — the acoustic-metric curvature is undefined within the zone (the metric itself is degenerate), but the underlying substrate has no divergent quantity.

A region where ED-gradients reach maximum viable value, where the acoustic metric loses representational validity, and where committed substrate participation accumulates without being able to propagate outward.

The continuum geometric description fails inside the zone, but the substrate description remains finite and well-defined throughout. There is no infinity anywhere in the substrate description. The “infinity” of classical GR singularities is a property of the failed continuum approximation, not of the underlying substrate.

This is the second piece of the no-paradoxes-because-no-smooth-edges framing: the interior at substrate level isn’t a singular point that information can be destroyed at. The substrate is finite, discrete, and well-defined throughout; what fails is the continuum geometric language.

5 Information Architecture and Evaporation

The third structural result is the substrate-level account of information flow at black holes. The argument runs through the committed-vs-uncommitted distinction, why information cannot cross the horizon, why entanglement can straddle it, the participation-re-routing mechanism for evaporation, and the explicit statement of why the standard paradoxes are not generated.

5.1 Information vs entanglement: substrate distinction

The key substrate distinction is between two structurally different forms of correlation:

Committed structure (information). A pattern in the substrate that has gone through P11 commitment — participation events have been irreversibly recorded in the local ED-history. Committed structure at point A can only be “read” by point B if there exists a participation pathway from A to B across which substrate events can propagate. This requires nonzero cross-bandwidth Γ_{cross} .

Uncommitted structure (entanglement). A correlation potential between two regions — the joint participation-amplitude structure prior to commitment. Uncommitted structure does not require ongoing cross-bandwidth between the two regions. It is set up by the prior participation history that produced the pair, and it persists as a correlation in joint outcomes whenever commitment events occur.

The structural difference:

Committed structure transmission requires nonzero Γ_{cross} at the surface separating sender and receiver.

Entanglement amplitude depends on the correlation structure laid down at pair-formation; once formed, it does not require ongoing cross-bandwidth to persist.

Schematically:

$$\Gamma_{\text{cross}} \rightarrow 0 \implies \text{committed structure blocked, but entanglement amplitude remains finite}$$

In standard QFT terms: a horizon blocks signaling but not pre-existing correlations. In framework terms: a horizon blocks committed structure transmission but not uncommitted structure persistence.

This is not a new claim — it is the substrate translation of the standard quantum-information no-signaling theorem applied at a horizon. What is new in the framework is that both sides — committed and uncommitted — have substrate-level definitions, so the no-signaling-but-correlations-persist behavior is derived rather than postulated.

5.2 Why information cannot cross a horizon

From §3, the decoupling surface is defined by exponential cross-bandwidth suppression. The argument:

Committed structure transmission across a surface Σ requires nonzero Γ_{cross} across Σ — without ongoing cross-bandwidth, no participation event on side A can register as a participation event on side B .

At the decoupling surface, Γ_{cross} is below hydrodynamic-window resolution. From the coarse-grained continuum perspective accessible to any external observer, the cross-bandwidth is effectively zero.

Therefore committed structure cannot cross the decoupling surface. Information, in the substrate sense, does not pass through a horizon.

This is not a causal prohibition. The framework has no fundamental light-cone primitive — light cones emerge as level sets of the acoustic metric, which is itself a coarse-grained reading. The prohibition against information crossing the horizon is a participation-bandwidth prohibition: the substrate channels that would carry committed structure have their bandwidth suppressed below resolution.

The distinction matters for two reasons. It does not require any global causal-structure argument; it follows from local bandwidth suppression. And it is consistent with finite- Γ_{cross} tunneling behavior near (but not beyond) the surface — the suppression is exponential, not a hard cutoff. This will be load-bearing for the evaporation mechanism below.

5.3 Evaporation as participation re-routing

The evaporation mechanism follows directly from the structure established above:

Horizon blocks committed structure. Uncommitted structure can straddle the surface.

Local participation gradients near the horizon support pair-creation-class events. The decoupling surface is where $|\nabla\rho| \cdot \ell_P^2/\rho_{\text{local}}$ is large. Large local gradients in the participation field are precisely the conditions under which uncommitted pair structures emerge — this is the substrate-level reading of vacuum pair production in a strong field, but here it is gradient-driven rather than externally-sourced.

One branch commits inward, the other commits outward. When commitment events occur on these straddling pair structures (as they generically do, via P11), the inward leg commits as part of the interior saturated zone (per §4), and the outward leg commits as part of the exterior radiation field. The pair was uncommitted at formation; its commitment events are local and irreversible.

This is participation re-routing. No committed structure crosses the horizon. Instead, the correlation structure that was laid down at pair-formation is now distributed across an interior commitment plus an exterior commitment, with the joint correlation preserved.

The outward branch inherits correlation structure from the interior branch via the joint amplitude that produced the pair. Globally, the correlation structure is preserved — no information is destroyed; it has been re-routed through pair-creation-class events at the decoupling surface, with one half ending up outside and one half ending up in the interior saturated zone.

What an external observer reads as “thermal Hawking radiation” is, in the framework, the coarse-grained statistics of the outward-committed legs of these straddling pair structures. The thermal character — and the spectrum, and the area-law-1/4 entropy — is not derived in this walkthrough. The substrate-level mechanism for evaporation is in hand: evaporation is participation re-routing, not information destruction. The full Hawking-spectrum derivation remains future work.

5.4 Why the framework has no information paradox

The standard information-loss paradox arises by combining four assumptions:

(a) Global unitarity of the full quantum state. (b) Global Cauchy-data evolution on a connected spacelike surface. (c) A geometric horizon as a sharp boundary. (d) Monogamy of entanglement enforced at that geometric boundary.

The framework imposes none of these assumptions at the substrate level:

No global unitarity requirement. The framework is local and irreversible at the substrate level (P11). Unitarity is a coarse-grained continuum property of the QM-emergence sector for closed sub-systems in the canonical hydrodynamic window. It is not a global constraint on the substrate. There is therefore no global-unitarity requirement that must be reconciled with horizon-blocked information.

No need for global Cauchy data. The framework commits events locally; there is no spacelike surface on which the entire universe’s state must be specified. The “information” question is answered locally, event by event, by P11.

No firewall. Firewalls arise from forcing entanglement-monogamy at a geometric surface where complementarity also requires preservation of effective-field-theory smoothness. In the framework, the horizon is a decoupling surface, not a geometric boundary. There is no requirement to enforce monogamy at it because the surface is a coarse-grained statistical feature of Γ_{cross} , not a knife-edge geometric locus.

No complementarity contradiction. Complementarity arises from asking whether information is “inside” or “outside” — a question that presupposes a sharp geometric boundary. In the framework the horizon is not a geometric boundary; the relevant question is whether

commitment events have occurred on a given side, and that question has a local answer for each event.

No information-loss paradox. Information never crosses the horizon (§5.2); committed structure inside the horizon remains finite (§4); evaporation re-routes correlation structure (§5.3). The exterior radiation carries correlation with the interior commitments via the joint amplitude structure of the pair-creation events. No committed structure is lost.

The structural point: the information paradox is not solved in the framework; it does not arise. The four assumptions that generate it are not present at the substrate level. They are continuum-level assumptions imposed in standard treatments because the underlying ontology is a global field on a fixed spacetime. The framework’s ontology is local participation events with irreversible commitment, and at that level the paradox-generating assumptions are not available to be made.

This is the third and most distinctive piece of the no-paradoxes-because-no-smooth-edges framing. The framework handles black-hole “paradoxes” by not generating them in the first place, rather than by resolving them within a paradox-generating framework.

6 Entropy: Form Forced, Coefficient Inherited

The fourth structural result is the area-law form of black-hole entropy, derived from substrate machinery, plus an honest verdict on the Bekenstein–Hawking $1/4$ coefficient.

6.1 The participation capacity of a horizon

Define the participation-capacity density $\chi(\rho, \nabla\rho)$ as the substrate-level density of viable commitment-channel slots per Planck-area patch on the decoupling surface Σ_H . By construction:

$\chi \rightarrow 1$ in the saturated regime (per §4): each Planck patch supports its full structural quota of commitment channels.

$\chi \rightarrow 0$ in the dilute regime: cross-bandwidth is plentiful but commitment-channel density is low.

The total participation capacity of Σ_H is:

$$C_H = \int_{\Sigma_H} (dA/\ell_P^2) \cdot \chi(\rho, \nabla\rho)$$

Two structural properties are load-bearing:

Area-extensivity. C_H scales with the integrated surface area, not with the enclosed volume, because the integrand is supported only on Σ_H . This is where the area-law form enters: it is forced by the fact that the decoupling surface is two-dimensional, not by any additional assumption.

Strong-gradient saturation. On a black-hole decoupling surface, the gradient condition that defines Σ_H is precisely the condition under which χ saturates: the same large $|\nabla\rho| \cdot \ell_P^2/\rho_{\text{local}}$ that suppresses cross-bandwidth also drives commitment-channel density to its substrate-imposed maximum. Therefore in the BH-relevant regime $\chi \approx 1$, and the capacity reduces to:

$$C_H \approx A/\ell_P^2$$

where A is the coarse-grained acoustic-metric area of Σ_H .

This recovers the area-law form at the substrate level: the count of commitment-channel slots on a black-hole horizon is A/ℓ_P^2 , with the leading coefficient set to unity by gradient saturation.

6.2 Entropy as count of viable commitment histories

The framework’s substrate-level definition of entropy is:

$$S = \log(\text{number of viable commitment histories compatible with the macroscopic constraint})$$

where “macroscopic constraint” is the participation capacity of Σ_H . A commitment history is a P11-irreversible record of which channels committed in which order, subject to V1 kernel finite-width constraints plus DCGT hydrodynamic-window resolution.

Each Planck-area patch on Σ_H supports a finite number g of distinct viable commitment motifs (the substrate’s local “alphabet” of commitment-channel configurations after coarse-graining). Patches commit independently up to the resolution of the hydrodynamic window; correlations across patches are sub-leading in the multi-scale expansion.

Therefore:

Total number of viable commitment histories: g^{C_H} .

Entropy:

$$S = C_H \cdot \log g \approx (A/\ell_P^2) \cdot \log g$$

The area-law form is now derived at the substrate level: $S \propto A/\ell_P^2$ follows from two-dimensional surface support, gradient saturation, and per-patch finite alphabet.

6.3 The 1/4 coefficient: inherited rather than derived

Bekenstein–Hawking entropy: $S_{BH} = A/(4\ell_P^2)$.

Framework result: $S_{ED} = (A/\ell_P^2) \cdot \log g$.

Form match: both are area-law with coefficient $1/\ell_P^2$ multiplied by a dimensionless number. The framework reproduces the area-law form structurally.

Coefficient comparison: the framework predicts $\log g$, with g a substrate-determined finite integer ≥ 2 . The Bekenstein–Hawking value $1/4$ corresponds to $\log g = 1/4$, i.e., $g = e^{1/4} \approx 1.284$.

This is not an integer, which is the structurally informative outcome. It tells us:

Naive per-patch-independent commitment-motif counting cannot reproduce $1/4$ exactly: a finite-alphabet log gives $\log g$ for integer $g \geq 2$, all of which exceed $1/4$.

To recover $1/4$, the per-patch independence assumption must be relaxed. Adjacent patches share commitment-history correlations through V1 kernel overlap, and the effective per-patch contribution is reduced from $\log g$ (independent) to a smaller correlated value.

The Bekenstein–Hawking coefficient is consistent with $\log g \approx 1/4$ once V1 kernel cross-patch correlations are accounted for — which is structurally plausible because the V1 kernel finite width is precisely what links adjacent patches. A closed-form derivation requires substrate-level state counting that is not currently in hand.

The honest verdict: the area-law form is derived; the $1/4$ coefficient is inherited rather than derived. This is the same structural pattern the framework uses elsewhere — form forced by substrate mechanics, value inherited from substrate constants. The $1/4$ coefficient is in the same category as the gauge couplings, the CKM matrix elements, and other parameters where the framework derives the structural form within which the constant appears but inherits the specific numerical value.

A substrate-level state-counting derivation of $\log g$ is a candidate future arc. It would require a closed-form V1-kernel motif count, which requires extending the current machinery beyond what’s in hand.

Observational distinguishability: the black-hole entropy coefficient is not directly measured. It is inferred from thermodynamic consistency with the Hawking temperature. Since the Hawking-spectrum derivation is also flagged as future work, there is no current observation that the framework's result disagrees with at the empirical level. The result is a structural-positive: form reproduced, coefficient identified as inherited, consistent with $1/4$ pending closed-form computation.

7 Wave-Black-Hole Scattering

The fifth structural result is the framework's account of how gravitational waves scatter off black holes. This is the most technical of the BH derivations and is presented compactly here; the central content is that the BHPT phase shift emerges as a global path-integrated invariant of a substrate-level minimal channel passing through the saturated-gradient region.

7.1 Setup

A minimal ED-channel (a gravitational wave, in continuum reading) is a substrate excitation characterized by low multiplicity and low anisotropy. Its global phase along a propagation trajectory $\gamma(\lambda)$ is:

$$\Phi[\gamma] = \int_{\gamma} \mathcal{A}(\rho, \nabla\rho, \text{anisotropy}) d\lambda$$

where $\mathcal{A}$ is the channel's substrate action density. After DCGT, this reduces in the asymptotic region to the standard kinetic-plus-dispersion terms of a wave on the acoustic metric.

A maximal ED-structure (a black hole) has high multiplicity, saturated gradients near Σ_H , and curvature-like participation structure given by SG-4 in the relevant weak-field regime.

7.2 The BHPT phase shift

The minimal channel's propagation kernel is distorted by the maximal structure's saturated gradients. Strong $|\nabla\rho|$ contributes to the acoustic-metric curvature, which appears in $\mathcal{A}$ as a curvature-coupling term beyond the flat-space free-wave action.

The channel's global phase accumulates a shift relative to its asymptotic free-wave value:

$$\delta_{\ell m} = \int_{\text{path}} \Delta\mathcal{A} d\lambda$$

where $\Delta\mathcal{A} = \mathcal{A}_{\text{full}} - \mathcal{A}_{\text{asymptotic}}$ is the curvature-induced excess action density along the trajectory.

The shift is global because the channel is minimal: the low-multiplicity, low-anisotropy structure means the shift collapses to a single scalar per partial-wave mode (ℓ, m) . The mode decomposition is inherited from the asymptotic axisymmetric structure of the BH background.

The shift is inevitable for any non-trivial channel trajectory through the strong-curvature region: gradient saturation near Σ_H guarantees $\Delta\mathcal{A} \neq 0$ along any trajectory that approaches the surface.

The S-matrix structure $S = e^{iN}$ is recovered: N is the integrated channel action, $\delta_{\ell m}$ are its partial-wave components, and unitarity of S on a per-channel basis follows from the reality of $\mathcal{A}$ in the exterior region (no committed structure leaks into the interior, so no probability is lost from the scattering channel).

7.3 Helicity preservation and flip

The channel’s polarization is encoded in its anisotropy tensor. Polarization transport along the trajectory is governed by how the anisotropy basis evolves under the background participation profile.

Helicity preservation (Schwarzschild-like). For axisymmetric, non-rotating backgrounds, the principal directions of the participation profile are aligned along the radial-tangential axes uniformly along the trajectory’s symmetry plane. The channel’s anisotropy tensor, parallel-transported along the trajectory, returns to itself up to an overall scalar phase. Helicity is preserved; the phase shift acts identically on both helicity states.

Helicity flip (Kerr-like). For backgrounds with frame-dragging — non-zero substrate-level vorticity ω_{FD} in the participation flow — the principal directions rotate along the trajectory. The accumulated twist is:

$$\Delta\phi_{\text{twist}} = \int \omega_{FD} d\lambda$$

The two circular-polarization eigenstates are no longer eigenstates of parallel transport; they couple via the twist. Helicity flip channels open with mixing strength tied to the spin parameter.

7.4 Kerr twist as integrated vorticity

Frame-dragging is a global rotation of the participation anisotropy basis. A spinning maximal structure carries non-zero substrate-level vorticity in its participation flow. The channel acquires a twist proportional to integrated vorticity. Because the twist depends on the alignment of the channel’s trajectory with the spin axis, scattering is no longer isotropic in the trajectory’s azimuthal direction. Channels with trajectories prograde to the spin acquire a different twist than retrograde channels. This is the substrate origin of the prograde/retrograde asymmetry in Kerr scattering.

Superradiance — energy extraction from a Kerr BH by certain scattering channels — corresponds to channels whose interaction with the rotating participation flow nets a positive energy transfer from background to channel. The mechanism is consistent with the information architecture: no committed structure is extracted from the interior; the energy transfer is mediated by the exterior participation flow’s vorticity acting on the channel. Full superradiance derivation is deferred.

What’s derived: the BHPT phase shift form as a global invariant per partial-wave mode, helicity preservation/flip from anisotropy-basis transport, Kerr twist from integrated frame-dragging vorticity. What’s inherited: specific numerical values of $\delta_{\ell m}$ conditional on the acoustic-metric scalar-tensor approximation matching GR Schwarzschild/Kerr in the relevant regime, and specific superradiance amplitudes.

8 The Architecture as a Whole

Stepping back, the structural shape of the BH arc is worth naming.

The QM walkthroughs each had specific load-bearing arguments concentrated in particular places. The gravity walkthrough had four distinct results composing into the galactic-dynamics chain. The arrow-of-time walkthrough had a single chain of forcing arguments. The BH arc has a different structural shape: one substrate-level condition supporting six different consequences.

The condition: $|\nabla\rho| \cdot \ell_P^2/\rho_{\text{local}} \gtrsim \log(R_{\text{cg}}/\ell_P)$.

What it does:

Defines the horizon as a decoupling surface (§3). Where the condition is met, cross-participation bandwidth is exponentially suppressed below DCGT resolution, and the surface separates regions that cannot exchange committed structure across it.

Marks the strong-curvature audit threshold (§4). Beyond this same condition, the acoustic-metric approximation breaks down and the continuum geometric description fails. The classical singular point is replaced by a saturated participation zone where substrate dynamics remain finite but geometric language is no longer applicable.

Sets the cross-bandwidth suppression that blocks committed-structure transmission (§5). The same condition that defines the horizon as a decoupling surface is what blocks information from crossing it.

Defines the gradient saturation that drives $\chi \rightarrow 1$ in entropy capacity (§6). The same condition that suppresses cross-bandwidth also drives commitment-channel density to its substrate-imposed maximum, giving the area-law form $S \propto A/\ell_P^2$.

Controls the strong-curvature scattering region (§7). The exterior region where the channel acquires its phase shift is where the condition approaches its critical value but doesn't yet exceed it; beyond the surface, no committed structure can scatter through.

One condition, six consequences. The architecture is internally consistent — every load-bearing piece traces to substrate primitives plus the same gradient-threshold condition. This is the structural unification that makes the framework's BH content distinctive.

The standard paradoxes do not arise because the four assumptions that generate them — global unitarity, global Cauchy data, geometric horizon as primitive boundary, monogamy at boundary — are not imposed at the substrate level. The horizon is a participation-bandwidth-collapse surface, not a smooth geometric primitive. The interior has no singularity. Information is locally committed and bandwidth-blocked from crossing the decoupling surface. Evaporation is participation re-routing through pair-creation events, with global correlation preserved.

The structural payoff: black-hole architecture in the framework is what falls out when discreteness, finite participation, and commitment irreversibility are pushed through to the strong-gradient regime. The smooth-horizon idealization that the standard paradoxes require simply isn't there. The substrate is discrete. The horizon isn't smooth. The paradoxes don't arise because the assumptions that generate them aren't present.

9 What's Forced, What's Inherited, What's Open

The framework is honest about what its current machinery delivers and what remains future work. The BH arc lands at a specific empirical position with specific gaps named.

Forced at substrate level (form):

- Horizon as decoupling surface from gradient steepening plus V1 finite-width kernel plus DCGT multi-scale expansion (§3).
- Universalization across BH, Rindler, cosmological, and acoustic horizons under one mechanism (§3.5).
- Geometric event horizon as continuum-emergent shadow of substrate decoupling surface; horizon teleology dissolved (§3.6).
- Absence of singularities from three substrate constraints jointly forbidding divergence (§4.1).
- Finite-thickness saturated participation zone replacing the classical singular point (§4.3).
- Information-blocking at the horizon as participation-bandwidth prohibition rather than causal prohibition (§5.2).

- Entanglement-straddling at the horizon as uncommitted structure persistence (§5.1).
- Evaporation as participation re-routing through pair-creation-class events at the decoupling surface (§5.3).
- No information paradox because four paradox-generating assumptions are not imposed at substrate level (§5.4).
- Area-law entropy form $S \propto A/\ell_P^2$ from gradient saturation plus per-patch finite alphabet plus two-dimensional surface support (§6.1–6.2).
- BHPT phase shift as global path-integrated channel-action invariant; helicity preservation/flip from anisotropy-basis transport; Kerr twist as integrated frame-dragging vorticity (§7).

Inherited at value layer:

- Specific gradient threshold $|\nabla\rho|_c$ (set by V1 profile plus DCGT scale separation).
- Numerical value of α (V1-profile coefficient in exponential suppression).
- Specific value of β_{crit} (DCGT scale separation plus V1 profile).
- Numerical value of ρ_{max} (substrate participation-density saturation scale).
- Specific size and shape of saturated zone for given BH mass.
- Bekenstein–Hawking 1/4 coefficient — area-law form is forced, but $\log g \approx 1/4$ requires V1 kernel cross-patch correlation calculations not currently in hand.
- Specific numerical values of BHPT spectra δ_{ℓ_m} (conditional on acoustic-metric matching GR Schwarzschild/Kerr in relevant regime).
- Specific superradiance amplitudes.

Open (future arc work):

Hawking spectrum derivation. The mechanism (asymmetric participation at saturated decoupling surface, pair-creation-class events) is in hand from §5. The explicit spectrum derivation matching $T_H = \kappa/(2\pi)$ requires V5 cross-chain correlation calculation that’s flagged as a separate technically-heavy follow-on arc. This is the most empirically significant open item — the framework currently does not predict the Hawking temperature, only the qualitative thermal-character of evaporation radiation.

Substrate-level state counting for $\log g$. Closed-form V1-kernel motif count would derive the 1/4 coefficient explicitly. Structurally similar to other anchor searches in the program — requires extending current machinery.

Full Kerr interior audit. The framework’s strong-curvature audit establishes interior saturated-zone structure for Schwarzschild-class BHs. Rotating BH interior architecture (ergosphere, Cauchy horizons, frame-dragging at saturation) requires extension.

Full constructive-rigorous proof of information preservation. The framework produces a structural-suggestive resolution of the information paradox — no paradox arises because paradox-generating assumptions aren’t imposed. It does not produce a constructive Streater–Wightman or OS-axiom-class proof.

Beyond acoustic-metric approximation. The framework operates within the acoustic-metric scalar-tensor approximation. Full GR remains speculative, not derived. Strong-curvature regimes beyond acoustic-metric extension are out of current scope.

10 What This Argument Establishes

The chain runs:

Primitives (micro-events, participation, chain stability landscape, decoupling surfaces, V1 finite-width kernel, P11 commitment irreversibility, participation-count bound) $\rightarrow$ DCGT machinery

(substrate-to-continuum bridge, hydrodynamic-window scale separation, multi-scale expansion) $\rightarrow$ critical gradient condition $|\nabla\rho| \cdot \ell_P^2/\rho_{\text{local}} \gtrsim \log(R_{\text{cg}}/\ell_P) \rightarrow$ six structural consequences (horizon formation, no singularities, information blocking, entanglement straddling, evaporation as participation re-routing, area-law entropy, BHPT phase shift) $\rightarrow$ no standard paradoxes generated because four paradox-generating assumptions not imposed.

Each consequence has its load-bearing argument worked out. The horizon is a participation-bandwidth-collapse surface, not a smooth geometric boundary. The interior is a finite-thickness saturated participation zone, not a singular point. Information is locally committed and bandwidth-blocked from crossing the decoupling surface. Evaporation is participation re-routing with global correlation preserved. Entropy follows the area-law form with the $1/4$ coefficient inherited rather than derived. Scattering produces the BHPT phase shift as a global path-integrated invariant.

The framework reproduces the architectural features of black-hole physics. It does not derive the Hawking spectrum. It does not produce the $1/4$ coefficient in closed form. It operates within the acoustic-metric scalar-tensor approximation rather than full GR. These are honestly named limitations, not hidden assumptions.

What the framework does that standard treatments do not: it shows that the standard paradoxes are not generated when the substrate is discrete and commits irreversibly. The information paradox, the firewall, the complementarity contradiction — all of them depend on structural assumptions (global unitarity, global Cauchy data, geometric horizon as primitive, monogamy at boundary) that the framework’s substrate ontology does not make. Without those assumptions, the paradoxes do not arise.

The framing your phrasing captured is exactly right: black holes don’t have paradoxes because black holes don’t have smooth edges, because the substrate is discrete. The smooth-horizon idealization that the paradoxes require simply isn’t there. The framework’s contribution is to make that explicit at every load-bearing step — to show, derivation by derivation, that horizon mechanism, interior structure, information flow, entropy form, and scattering structure all trace to substrate primitives plus one critical-gradient condition, with no place for the smooth-boundary idealization to enter.

For black-hole architecture specifically, the structural case is closed at the form level. The horizon mechanism is forced. The absence of singularities is forced. The information architecture is forced. The area-law entropy form is forced. The scattering structure is forced. The unification of all five under one substrate condition is forced. The $1/4$ coefficient and the Hawking spectrum are honestly flagged as inherited or open.

Whether the substrate commitments are right is the load-bearing question, as in every walk-through. The framework stands or falls on whether discreteness, finite participation bandwidth, commitment irreversibility, and the rest of the substrate ontology are the correct foundational concepts. The empirical exposure of the BH content lives downstream — in any future test of substrate-level black-hole physics that distinguishes the framework from standard treatments. The most striking empirical signature, if it can be developed, would be the Hawking-spectrum derivation matching or differing from the standard $\kappa/(2\pi)$ result. That’s open work.

For the architectural picture — what black holes are, why they don’t have singularities, why information flow at horizons doesn’t generate paradoxes, why the area-law entropy form is what it is — the framework’s substrate-level account is in hand. The smooth-horizon idealization that has generated forty years of paradox literature is replaced by a discrete, bandwidth-collapse-driven, participation-network description in which the paradoxes don’t arise because the assumptions that generate them aren’t present.

That’s the framework’s contribution to black-hole physics. Not paradox resolution within

a paradox-generating framework, but architectural completeness in a framework where the paradoxes don't get generated. The discreteness of the substrate, pushed through to the strong-gradient regime, is what makes black holes paradox-free at the structural level.

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